

EC-341 Digital System Design

BS(CE)-2021

LAB REPORT # 06



Submitted By:

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	Presentation (Format(0.5)+ Title (0.5)+ Conclusion(1) +Procedure(2) +Objective(1))	Calculation/ Coding	Observation/ Program Results	Total
Total	5	5	5	15
Obtained				

Lab Instructor.
Engr. Fasih Ahmad

Experiment #06

Analysis of Implementation of Combinational Circuits

Using Different Approaches

Decoders and Multiplexers

Objective

- **Combinational Circuits**
- **Implementation using Discrete Gates**
- **Implementation using Decoders**
- **Implementation using Multiplexers**
- **Lab Tasks**

Software Used

- Vivado

Tasks

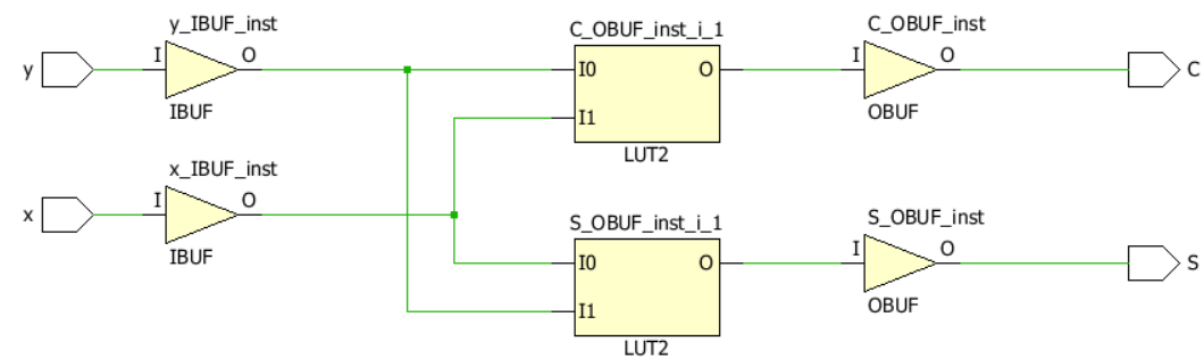
Task 1

Half adder Gate-Level

```

23 module Hadd(
24     input x,
25     input y,
26     output S,
27     output C
28 );
29     xor(S,x,y);
30     and(C,x,y);
31 endmodule
32

```



```

29 1. Summary
30 -----
31
32 +-----+-----+-----+
33 | Total On-Chip Power (W) | 0.773 |
34 | Dynamic (W)             | 0.665 |
35 | Device Static (W)       | 0.108 |
36 | Effective TJA (C/W)    | 11.5  |
37 | Max Ambient (C)        | 76.1  |
38 | Junction Temperature (C) | 33.9  |
39 | Confidence Level        | Low   |
40 | Setting File            | ---   |
41 | Simulation Activity File | ---   |
42 | Design Nets Matched    | NA    |
43 +-----+-----+-----+
44
45

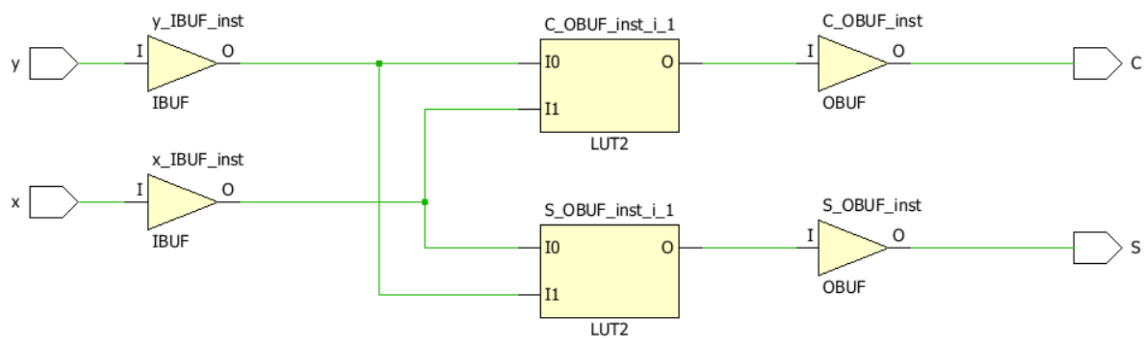
```

Half adder Behavior level

```

23 module Hadd(
24     input x,
25     input y,
26     output S,
27     output C
28 );
29 reg S, C;
30 always @(*) begin
31     S = x ^ y;
32     C = x & y;
33 end
34 endmodule
35

```



	Cell	Count
1	LUT2	4
2	LDC	1
3	LDCP	1
4	IBUF	2
5	OBUF	2

Report Instance Areas:

	Instance	Module	Cells
1	top		10

```

29 1. Summary
30 -----
31
32 +-----+-----+
33 | Total On-Chip Power (W) | 0.773 |
34 | Dynamic (W)             | 0.665 |
35 | Device Static (W)       | 0.108 |
36 | Effective TJA (C/W)     | 11.5  |
37 | Max Ambient (C)         | 76.1  |
38 | Junction Temperature (C) | 33.9  |
39 | Confidence Level         | Low   |
40 | Setting File             | ---   |
41 | Simulation Activity File | ---   |
42 | Design Nets Matched     | NA    |
43 +-----+-----+

```

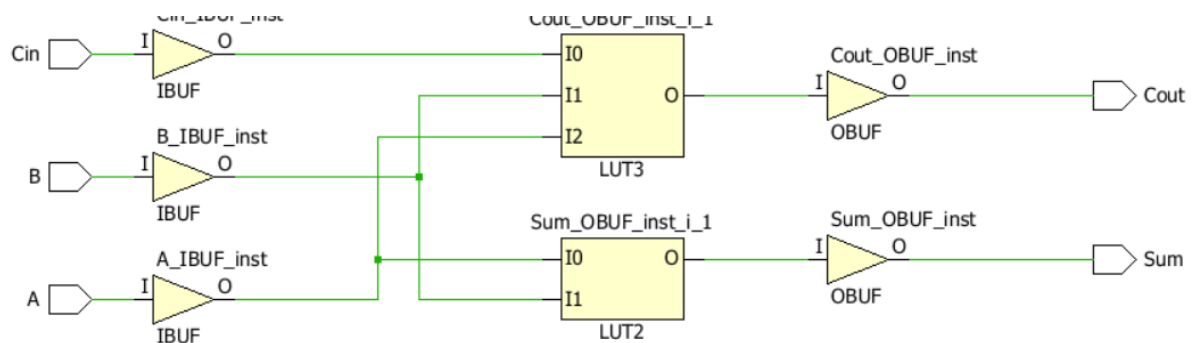
Full Adder

Gate level Implementation

```

23 module FullAdder(
24     input A,
25     input B,
26     input Cin,
27     output Sum,
28     output Cout
29 );
30
31 wire w1, w2, w3;
32
33 xor xor1(Sum, A, B);
34 xor xor2(w1, Cin, Sum);
35 and and1(w2, A, B);
36 and and2(w3, Cin, Sum);
37 or or1(Cout, w1, w2, w3);
38
39 endmodule
40
41

```



209	Report Cell Usage:	29	1. Summary
210	+-----+-----+-----+	30	-----
211	Cell Count	31	
212	+-----+-----+-----+	32	+-----+-----+
213	1 LUT2 1	33	Total On-Chip Power (W) 0.712
214	2 LUT3 1	34	Dynamic (W) 0.604
215	3 IBUF 3	35	Device Static (W) 0.108
216	4 OBUF 2	36	Effective TJA (C/W) 11.5
217	+-----+-----+-----+	37	Max Ambient (C) 76.8
218		38	Junction Temperature (C) 33.2
219	Report Instance Areas:	39	Confidence Level Low
220	+-----+-----+-----+	40	Setting File ---
221	Instance Module Cells	41	Simulation Activity File ---
222	+-----+-----+-----+	42	Design Nets Matched NA
223	1 top 7	43	+-----+-----+
224	+-----+-----+-----+	44	
225	-----		

Full Adder Behavior Level

```

45     else begin
46         Sum = 0;
47         Cout = 1;
48     end
49 end
50 else begin
51     if (A == 0 && B == 0) begin
52         Sum = 1;
53         Cout = 0;
54     end
55     else if (A == 0 && B == 1) begin
56         Sum = 0;
57         Cout = 1;
58     end
59     else if (A == 1 && B == 0) begin
60         Sum = 0;
61         Cout = 1;
62     end
63     else begin
64         Sum = 1;
65         Cout = 1;
66     end
67 end
68 end
69 endmodule

```

```

23 module FullAdder(
24     input A,
25     input B,
26     input Cin,
27     output reg Sum,
28     output reg Cout
29 );
30
31 always @* begin
32     if (Cin == 0) begin
33         if (A == 0 && B == 0) begin
34             Sum = 0;
35             Cout = 0;
36         end
37         else if (A == 0 && B == 1) begin
38             Sum = 1;
39             Cout = 0;
40         end
41         else if (A == 1 && B == 0) begin
42             Sum = 1;
43             Cout = 0;
44         end

```

```

208
209 Report Cell Usage:
210 +-----+-----+-----+
211 |      |Cell |Count |
212 +-----+-----+-----+
213 |1     |LUT3 |    2|
214 |2     |IBUF |    3|
215 |3     |OBUF |    2|
216 +-----+-----+-----+
217
218 Report Instance Areas:
219 +-----+-----+-----+-----+
220 |      |Instance |Module |Cells |
221 +-----+-----+-----+-----+
222 |1     |top      |      |    7|
223 +-----+-----+-----+-----+
=224 -----

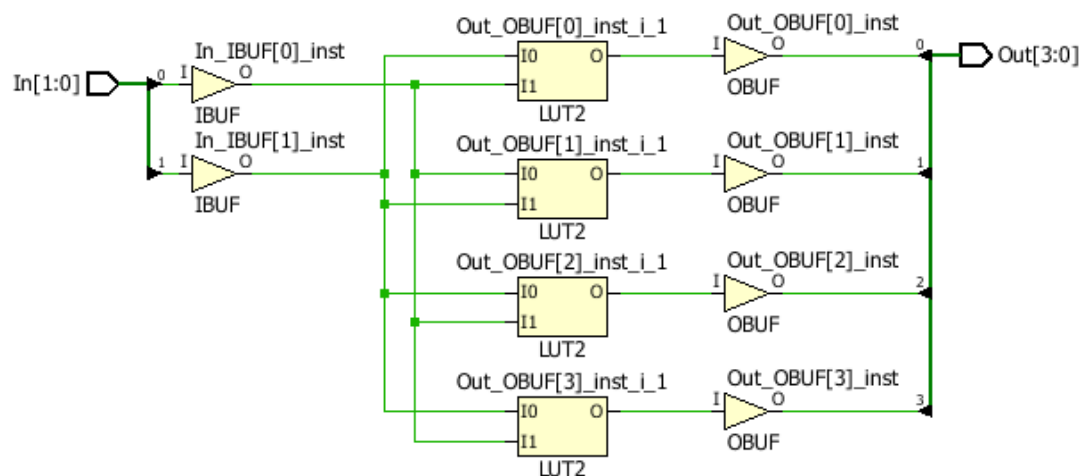
```

Task 2

```

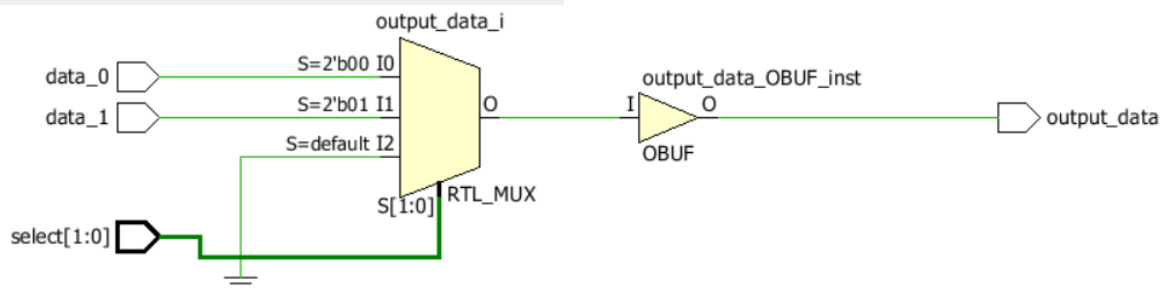
22 module Decoder(
23     input [1:0] input_code,
24     output reg [3:0] output_signals
25 );
26
27 always @* begin
28     case(input_code)
29         2'b00: output_signals = 4'b0001;
30         2'b01: output_signals = 4'b0010;
31         2'b10: output_signals = 4'b0100;
32         2'b11: output_signals = 4'b1000;
33         default: output_signals = 4'b0000;
34     endcase
35 end
36
37 endmodule
38

```



Task 3

```
23 module mux(  
24     input [1:0] select,  
25     input data_0,  
26     input data_1,  
27     output reg output_data  
28 );  
29  
30 always @* begin  
31     case(select)  
32         2'b00: output_data = data_0;  
33         2'b01: output_data = data_1;  
34         default: output_data = 1'b0;  
35     endcase  
36 end  
37  
38 endmodule
```



209 Report Cell Usage:

210	Cell	Count
211		
212		
213	LUT4	1
214	IBUF	4
215	OBUF	1
216		

217

218 Report Instance Areas:

219	Instance	Module	Cells
220			
221			
222	1	top	6
223			

Conclusion

In this lab we implemented different combinational circuits like half adder, full adder, decoder and mux.